

Medical Terminology — Word Parts

Formas combinantes, prefijos y sufijos · referencia de estudio · Inglés Médico I · Semana 2

Building & combining rules (high-yield)

- A term is built from **word root** + (combining vowel) + **suffix**, sometimes with a **prefix** in front.
- Every term **must** have a suffix; not every term has a prefix.
- The combining vowel is almost always **o** (sometimes **i**). A root written with its vowel = **combining form** (e.g. **cardi/o**).
- **Root + suffix**: suffix begins with a **vowel** → drop the vowel (**arthr + itis = arthritis**). Suffix begins with a **consonant** → keep it (**arthr/o + scope = arthroscope**).
- **Root + root**: keep the combining vowel even if the next root starts with a vowel (**gastr/o + enter/o + itis = gastroenteritis**).

Combining forms

bi/o	life	biology (study of life)	immun/o	immunity	immunology (study of immunity)
carcin/o	cancer	carcinoma (cancerous tumor)	laryng/o	larynx	laryngeal (pertaining to the voice box)
cardi/o	heart	cardiac (pertaining to the heart)	nephr/o	kidney	nephromegaly (enlarged kidney)
chem/o	chemical	chemotherapy (treatment with chemicals)	neur/o	nerve	neural (pertaining to a nerve)
cis/o	to cut	incision (process of cutting into)	ophthalm/o	eye	ophthalmic (pertaining to the eye)
dermat/o	skin	dermatology (study of the skin)	ot/o	ear	otic (pertaining to the ear)
enter/o	small intestine	enteric (pertaining to the small intestine)	path/o	disease	pathology (study of disease)
gastr/o	stomach	gastric (pertaining to the stomach)	pulmon/o	lung	pulmonary (pertaining to the lungs)
gynec/o	female	gynecology (study of females)	rhin/o	nose	rhinoplasty (surgical repair of the nose)
hemat/o	blood	hematic (pertaining to the blood)			

Common prefixes

a-	without	aphasia (without speech)	in-	not; inward	infertility (not fertile); inhalation (breathe in)
an-	without	anoxia (without oxygen)	inter-	between	intervertebral (between the vertebrae)
anti-	against	antibiotic (against life)	intra-	within	intravenous (within a vein)
auto-	self	autograft (graft from one's own body)	macro-	large	macrotia (having large ears)
brady-	slow	bradycardia (slow heartbeat)	micro-	small	microtia (having small ears)
de-	without	depigmentation (without pigment)	neo-	new	neonatology (study of the newborn)
dys-	painful; difficult; abnormal	dysuria (painful urination); dyspnea (difficulty breathing)	para-	beside; abnormal; two like parts	paranasal (beside the nose); paraplegia
endo-	within; inner	endoscope (view within); endocardium (inner lining of heart)	per-	through	percutaneous (through the skin)
epi-	above	epigastric (above the stomach)	peri-	around	pericardial (around the heart)
eu-	normal	eupnea (normal breathing)	post-	after	postpartum (after birth)
ex-	outward	exostosis (projecting bone)	pre-	before	preoperative (before surgery)
extra-	outside of	extracorporeal (outside of the body)	pro-	before	prolactin (before milk)
hetero-	different	heterograft (graft from another species)	pseudo-	false	pseudocyesis (false pregnancy)
homo-	same	homograft (graft from the same species)	re-	again	reinfection (to infect again)
hyper-	excessive	hypertrophy (excessive development)	retro-	backward; behind	retrograde; retroperitoneal (behind peritoneum)
hypo-	below; insufficient	hypodermic (below the skin); hypoglycemia	sub-	under	subcutaneous (under the skin)
			tachy-	fast	tachycardia (fast heartbeat)
			trans-	across	transurethral (across the urethra)
			ultra-	beyond	ultrasound (beyond sound)
			un-	not	unconscious (not conscious)

Number prefixes

bi-	two	bilateral (two sides)	pan-	all	pansinusitis (all the sinuses)
hemi-	half	hemiplegia (paralysis of half the body)	poly-	many	polymyositis (many muscles)
mono-	one	monoplegia (paralysis of one extremity)	quadri-	four	quadriplegia (all four limbs)
multi-	many	multigravida (pregnant more than once)	semi-	partial	semiconscious (partially conscious)
nulli-	none	nulligravida (no pregnancies)	tetra-	four	tetraplegia (all four limbs)
			tri-	three	triceps (muscle with three heads)

Ojo (confusiones de examen): mono/uni = uno · bi = dos · tri = tres · quadri/tetra = cuatro · hemi/semi = mitad/parcial · poly/multi = muchos · nulli = ninguno · pan = todo.

Common suffixes

-algia	pain	gastralgia (stomach pain)	-opsy	view of	biopsy (view of life)
-cele	protrusion	cystocele (protrusion of bladder)	-osis	abnormal condition	cyanosis (condition of being blue)
-cyte	cell	erythrocyte (red cell)	-pathy	disease	myopathy (muscle disease)
-dynia	pain	cardiodynia (heart pain)	-plasm	formation	neoplasm (new formation)
-ectasis	dilation	bronchiectasis (dilated bronchi)	-plegia	paralysis	laryngoplegia (paralysis of larynx)
-gen	that which produces	pathogen (produces disease)	-ptosis	drooping	blepharoptosis (drooping eyelid)
-genic	producing	carcinogenic (cancer producing)	-rrhage	excessive, abnormal flow	hemorrhage (excessive bleeding)
-ia	state, condition	bradycardia (condition of slow heart)	-rrhagia	abnormal flow condition	cystorrhagia (abnormal flow from bladder)
-iasis	abnormal condition	lithiasis (abnormal condition of stones)	-rrhea	discharge	rhinorrhea (discharge from the nose)
-ism	state of	hypothyroidism (state of low thyroid)	-rrhexis	rupture	hysterorrhexis (ruptured uterus)
-itis	inflammation	dermatitis (inflammation of skin)	-sclerosis	hardening	arteriosclerosis (hardening of an artery)
-logist	one who studies	cardiologist (studies the heart)	-stenosis	narrowing	angiostenosis (narrowing of a vessel)
-logy	study of	cardiology (study of the heart)	-therapy	treatment	chemotherapy (treatment with chemicals)
-lytic	destruction	thrombolytic (clot destruction)	-trophy	development	hypertrophy (excessive development)
-malacia	abnormal softening	chondromalacia (cartilage softening)			
-megaly	enlarged	cardiomegaly (enlarged heart)			
-oma	tumor, mass	carcinoma (cancerous tumor)			

Los 4 "-rrh-" que se confunden: -rrhage (flujo excesivo) · -rrhagia (condición de flujo) · -rrhea (secreción) · -rrhexis (ruptura).

Adjective suffixes — "pertaining to"

-ac	pertaining to	cardiac (the heart)	-ile	pertaining to	penile (the penis)
-al	pertaining to	duodenal (the duodenum)	-ine	pertaining to	uterine (the uterus)
-an	pertaining to	ovarian (the ovary)	-ior	pertaining to	superior (above)
-ar	pertaining to	ventricular (a ventricle)	-nic	pertaining to	embryonic (an embryo)
-ary	pertaining to	pulmonary (the lungs)	-ory	pertaining to	auditory (hearing)
-atic	pertaining to	lymphatic (lymph)	-ose	pertaining to	adipose (fat)
-eal	pertaining to	esophageal (the esophagus)	-ous	pertaining to	intravenous (within a vein)
-iac	pertaining to	chondriac (cartilage)	-tic	pertaining to	acoustic (hearing)
-ic	pertaining to	gastric (the stomach)			

Surgical suffixes

-centesis	puncture to withdraw fluid	arthrocentesis (from a joint)			thoracotomy (cutting into the chest)
-ectomy	surgical removal	gastrectomy (remove the stomach)	-pexy	surgical fixation	nephropexy (fixation of a kidney)
-ostomy	surgically create an opening	colostomy (opening for the colon)	-plasty	surgical repair	dermatoplasty (repair of the skin)
			-rrhaphy	to suture	myorrhaphy (suture muscle)
-otomy	cutting into		-tome	instrument to cut	dermatome (instrument to cut skin)

No confundir: -ostomy (crear una abertura) · -otomy (solo cortar) · -ectomy (extirpar).

Procedural suffixes

-gram	record or picture	electrocardiogram (record of heart's electricity)	-scope	instrument for viewing	gastroscope (view stomach)
-graphy	process of recording	electrocardiography	-scopic	pertaining to visual exam	endoscopic
-meter	instrument for measuring	audiometer (measure hearing)	-scopy	process of visually examining	gastroscopy (examine the stomach)
-metry	process of measuring	audiometry			

-gram = el registro · -graphy = el proceso. Igual: -meter (aparato) vs -metry (proceso); -scope (aparato) vs -scopy (proceso).